SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1019	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	20/10/2026	**Duration:**	60 minutes

Code of Conduct

I YAMUNA K / 192511013 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: *YAMUNA K*

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

Documentation (15%)	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

ANNEXURE A – SCENARIO-BASED REQUIREMENT ANALYSIS

Project Title: Smart Parking Management System
Domain: Transport and Parking Management

1. Study and Analyze the Scenario

1.1 Introduction

The Smart Parking Management System is a software application designed to simplify parking management in malls, colleges, campuses, and other crowded places. Drivers often waste time searching for available parking spaces, leading to traffic congestion and inconvenience.

The proposed system provides real-time parking slot availability, advance booking, automated entry and exit billing, and occupancy analytics. It helps drivers find parking spaces easily and enables operators to manage parking facilities efficiently.

1.2 Problem Description

Traditional parking systems depend on manual monitoring and do not provide accurate information about available parking slots. Drivers must search for empty spaces, resulting in wasted time, fuel consumption, and traffic congestion.

Manual entry and exit procedures also lead to billing errors and long queues. Parking operators face difficulties in monitoring occupancy and analyzing parking usage.

Therefore, a Smart Parking Management System is required to automate parking operations, improve slot utilization, reduce waiting time, and provide accurate parking information.

1.3 Business Domain

The system belongs to the Transport and Parking Management domain. It uses software, sensors, and digital payment facilities to improve parking operations in public and private parking areas.

1.4 Stakeholders

Stakeholder	Role
Drivers / Users	View available slots, book parking, and pay parking fees.
Parking Operators	Monitor parking slots, bookings, occupancy, and revenue.
System Administrator	Manage users, parking locations, permissions, and system settings.
Payment Gateway	Process digital parking payments securely.
Sensor / IoT System	Detect vehicle presence and update slot availability.
Management	Analyze parking usage, revenue, and occupancy reports.

1.5 Existing Challenges

1. Drivers spend excessive time finding available parking spaces.
2. Real-time information about vacant slots is unavailable or inaccurate.
3. Manual parking entry, exit, and billing create delays.
4. Double booking and improper slot allocation may occur.
5. Operators lack detailed occupancy and revenue reports.

1.6 Objectives of the System

1. Display real-time availability of parking slots.
2. Allow drivers to reserve parking slots in advance.
3. Automate vehicle entry, exit, and parking fee calculation.
4. Support secure digital payments and booking confirmation.
5. Generate occupancy and revenue analytics for parking operators.

2. Functional Requirements

Functional requirements describe the operations that the system must perform.

ID	Functional Requirement	Description
FR-01	User Registration and Login	The system shall allow users and operators to log in securely.
FR-02	Parking Location Search	The system shall allow drivers to search for parking locations.
FR-03	Real-Time Slot Availability	The system shall display available, occupied, and reserved slots.
FR-04	Advance Booking	The system shall allow drivers to reserve available slots for a selected time.
FR-05	Booking Confirmation	The system shall generate a unique booking ID and confirmation.
FR-06	Booking Cancellation	The system shall allow cancellation according to the parking policy.
FR-07	Automated Entry	The system shall verify a valid booking or parking authorization at entry.
FR-08	Automated Exit	The system shall record vehicle exit time and release the parking slot.
FR-09	Automatic Billing	The system shall calculate parking fees based on duration and applicable rates.
FR-10	Digital Payment	The system shall support payment through an integrated payment gateway.
FR-11	Notifications	The system shall notify users about booking confirmation, cancellation, and payment status.
FR-12	Slot Management	Operators shall be able to configure slots, parking rates, and availability.
FR-13	Occupancy Analytics	The system shall generate reports about occupied slots, available slots, and usage trends.
FR-14	Revenue Reports	The system shall generate parking revenue and transaction reports.
FR-15	Administration	The administrator shall manage user accounts, roles, locations, and system settings.

3. Non-Functional Requirements

Non-functional requirements define the quality, performance, and operational standards of the system.

ID	Requirement	Description
NFR-01	Performance	Slot availability and booking information should normally load within 3 seconds under expected operating conditions.
NFR-02	Security	Passwords shall be securely stored, and access shall be controlled using authentication and role-based authorization.
NFR-03	Reliability	The system shall maintain consistent booking and billing records and prevent double booking.
NFR-04	Usability	The interface shall be simple, clear, and easy to use for drivers and operators.
NFR-05	Scalability	The system shall support additional parking locations, slots, and users as demand increases.
NFR-06	Maintainability	The software shall be modular so that updates and bug fixes can be performed easily.

NFR-07	Availability	The system should target 99.5% monthly availability, excluding planned maintenance.
NFR-08	Data Integrity	Booking, vehicle entry/exit, and payment records shall remain accurate and consistent.
NFR-09	Compatibility	The application should work on commonly used web browsers and mobile devices.
NFR-10	Privacy	Personal, vehicle, and transaction information shall be protected and accessed only for authorized purposes.

4. Actors and Use Cases

4.1 Primary and Secondary Actors

Primary actors

1. Driver: Searches for slots, books parking, checks booking status, and pays fees.
2. Parking Operator: Manages parking slots, monitors occupancy, and views reports.
3. System Administrator: Manages accounts, permissions, and system configurations.

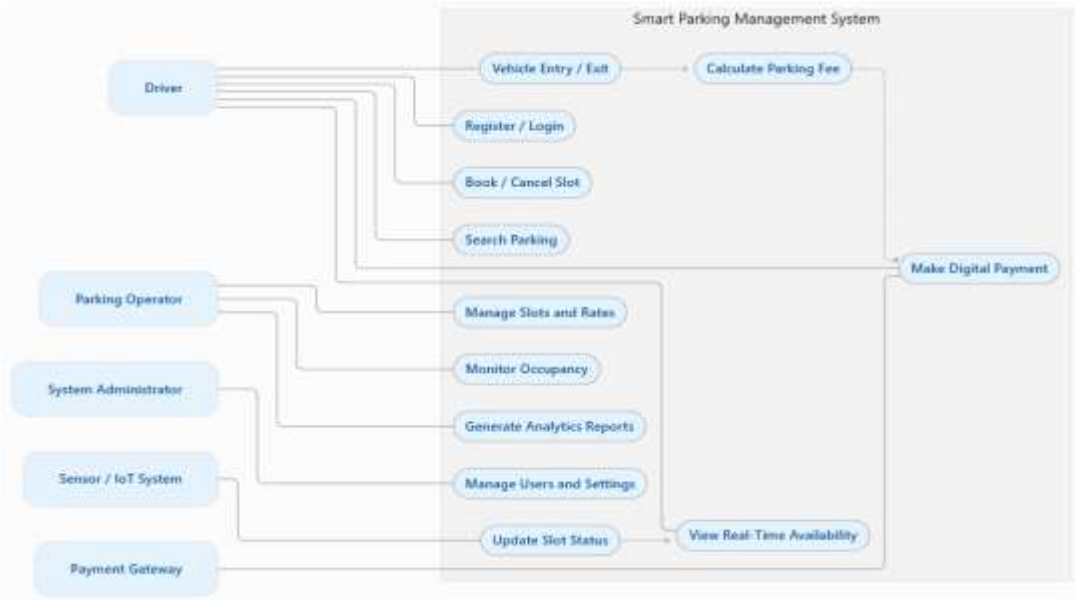
Secondary actors

4. Sensor / IoT System: Sends vehicle-presence and slot-occupancy information.
5. Payment Gateway: Processes payments and returns transaction status.

4.2 Major Use Cases

Actor	Use Cases
Driver	Register/Login, Search Parking, View Availability, Book Slot, Cancel Booking, Enter/Exit Parking, Pay Fee, View Booking Status
Parking Operator	Manage Slots, Update Rates, Monitor Occupancy, View Bookings, Generate Reports
System Administrator	Manage Users, Assign Roles, Manage Locations, Configure System
Sensor / IoT System	Detect Vehicle Presence, Update Slot Status
Payment Gateway	Process Payment, Return Payment Status

4.3 UML Use Case Diagram



Note: This shows the major system interactions. The dashed arrows indicate supporting relationships; in a detailed UML use case diagram, these may be represented using appropriate <<include>> relationships where applicable.

5. Software Requirement Specification (SRS)

5.1 Purpose

The purpose of this SRS is to define the functional and non-functional requirements of the Smart Parking Management System. It serves as a reference for developers, testers, operators, and stakeholders during system development and testing.

5.2 Scope of the System

The system provides a centralized platform for managing parking facilities.
Included in the scope:

- Real-time parking slot monitoring.
- Advance booking and cancellation.
- Automated vehicle entry and exit recording.
- Parking fee calculation and digital payments.
- Occupancy, usage, and revenue analytics.

Outside the initial scope:

- Construction of new parking facilities.
- Physical traffic management outside the parking area.
- Vehicle repair and maintenance services.
- Integration with every third-party parking platform.

5.3 Use Case Descriptions

Use Case 1: View Real-Time Slot Availability

Item	**Description**
Actor	Driver
Precondition	The system and parking availability service are operational.
Main Flow	Driver selects a parking location → System retrieves slot information → System displays available, occupied, and reserved slots.
Alternative Flow	If live data is unavailable, the system displays an update error and indicates that availability could not be verified.
Postcondition	Driver receives the latest available slot information provided by the system.

Use Case 2: Advance Parking Booking

Item	**Description**
Actor	Driver
Precondition	Driver is logged in and a suitable slot is available.
Main Flow	Driver selects location and time → System checks availability → Driver confirms booking → System reserves the slot and generates a booking ID.
Alternative Flow	If the slot becomes unavailable, the system asks the driver to select another slot.
Postcondition	A confirmed booking is stored and the slot is reserved for the specified period.

Use Case 3: Automated Entry and Exit Billing

Item	**Description**
Actor	Driver, Sensor / IoT System
Precondition	A valid booking or authorized parking session exists.
Main Flow	Vehicle enters → System records entry time → Vehicle exits → System records exit time → System calculates the fee → Driver completes payment.

Alternative Flow	If vehicle detection or payment processing fails, the system records the error and allows an authorized manual verification process.
Postcondition	The parking session and billing record are updated, and the slot becomes available after confirmed exit.

Use Case 4: Occupancy Analytics

Item	**Description**
Actor	Parking Operator
Precondition	The operator is authenticated and parking records are available.
Main Flow	Operator selects a date range → System collects parking data → System calculates occupancy and revenue → System displays reports.
Alternative Flow	If data is unavailable, the system displays an appropriate message.
Postcondition	The operator receives the requested analytics report.

5.4 Data Requirements

The system shall store and maintain the following information:

Data Entity	**Important Attributes**
User	User ID, name, contact details, password hash, role
Parking Location	Location ID, name, address, total capacity
Parking Slot	Slot ID, location ID, slot type, current status
Booking	Booking ID, user ID, slot ID, start time, end time, status
Vehicle	Vehicle ID, registration number, user ID
Parking Session	Session ID, vehicle ID, entry time, exit time
Payment	Payment ID, booking/session ID, amount, method, status
Analytics	Date, total slots, occupied slots, occupancy rate, revenue

The system shall maintain relationships between users, bookings, parking slots, parking sessions, and payments. Sensitive information shall be protected, and transaction records shall be retained according to applicable policies.

5.5 External Interface Requirements

1. User Interface
   - Simple web and mobile interfaces.
   - Search bar for parking locations.
   - Dashboard showing parking availability.
   - Booking, payment, and cancellation screens.
   - Operator dashboard with reports and charts.
2. Hardware Interface
   - Parking sensors to detect vehicle presence.
   - Entry and exit cameras or RFID readers, if installed.
   - Barrier gates or access-control devices, where supported.
3. Software Interface
   - Database for storing parking and transaction information.
   - Payment gateway for digital payments.
   - Notification service for booking and payment updates.
   - APIs for communication with sensors and access-control devices.
4. Communication Interface
   - Secure HTTPS communication between clients and servers.
   - Authenticated communication between the backend and supported devices.
   - Reliable data exchange between the system and external services.

5.6 System Features

1. Smart Slot Monitoring: Displays live parking availability.
2. Online Reservation: Allows advance booking and cancellation.
3. Automated Parking Operations: Records entry and exit and updates slot status.
4. Smart Billing: Calculates parking charges based on the configured tariff.
5. Analytics Dashboard: Provides occupancy trends, parking usage, and revenue reports.
6. Secure Administration: Controls access to user, slot, and system management functions.

6. Assumptions and Constraints

6.1 Assumptions

1. Drivers have access to a smartphone or web browser.
2. Parking locations provide accurate slot information through sensors or authorized updates.
3. Internet connectivity is available during normal system operation.
4. A supported payment gateway is available for digital transactions.
5. Parking rates and reservation policies are provided by the parking operator.
6. Users provide valid registration and vehicle information.

6.2 Constraints

Constraint Type	Description
Technical	Integration depends on compatible sensors, cameras, RFID readers, and APIs.
Operational	Parking staff must maintain equipment and handle exceptional cases.
Economic	Sensor installation, cloud hosting, software development, and maintenance require a budget.
Legal	The system must follow applicable data protection, privacy, payment, and record-retention requirements.
Time-related	Development and testing must be completed within the assigned project deadline.
Network	Internet or device failures may delay live availability updates and payment confirmation.

7. Requirement Completeness and Consistency Validation

Requirement validation ensures that the specification is complete, clear, consistent, and testable.

7.1 Completeness Validation

Validation Check	Result
All major stakeholders identified	Satisfied
Functional requirements cover booking, monitoring, billing, and analytics	Satisfied
Non-functional requirements cover performance, security, reliability, and usability	Satisfied
Actors and their use cases identified	Satisfied
Data and external interfaces specified	Satisfied
Assumptions and constraints documented	Satisfied
Alternative flows and error handling considered	Satisfied

7.2 Consistency and Ambiguity Validation

1. Availability consistency: Slot availability must be updated after confirmed bookings, vehicle entry, cancellation, and exit.
2. Double-booking prevention: The system must prevent two confirmed bookings from reserving the same slot for overlapping periods.
3. Billing consistency: Parking fees must be calculated using the configured tariff, recorded entry and exit times, and applicable rounding rules.
4. Payment consistency: A slot or booking must not be marked as paid until the payment gateway confirms the transaction.

5. Security consistency: Only authorized operators and administrators may modify parking rates, users, and system settings.
6. Data consistency: Booking, parking session, and payment records must use unique identifiers and maintain valid relationships.

7.3 Improvements to Enhance Requirement Quality

- Specify the expected maximum response time and concurrent-user capacity.
- Define how often sensors must update slot availability.
- Establish booking expiration, cancellation, and refund policies.
- Define tariff rules, grace periods, and fee-rounding methods.
- Specify recovery procedures for network, sensor, and payment failures.
- Define data backup, recovery, privacy, and retention policies.
- Prepare test cases for each functional requirement and measurable non-functional requirement.

7.4 Final Validation Statement

The proposed SRS covers the main requirements of the Smart Parking Management System, including stakeholders, system objectives, functional and non-functional requirements, actors, use cases, data, interfaces, assumptions, and constraints. The validation identifies important issues such as double booking, inaccurate slot updates, billing errors, and payment failures.

The specification is suitable as a baseline for design, implementation, and testing. Before deployment, the proposed response-time targets, availability targets, sensor-update frequency, and parking policies should be confirmed with the relevant stakeholders.

8. Conclusion

The Smart Parking Management System provides an efficient solution to common parking problems in malls and campuses. It reduces the time spent searching for parking spaces, supports advance reservations, automates entry and exit billing, and provides useful occupancy analytics.

A well-defined SRS helps ensure that the system meets user expectations, protects transaction data, and operates reliably. With proper implementation and validation, the proposed system can improve parking utilization, reduce congestion, and simplify parking management.